STATISTICAL LEARNING FOR AUTOMATION SYSTEMS

Model Identification and Data Analysis – Part 1

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The lecturer

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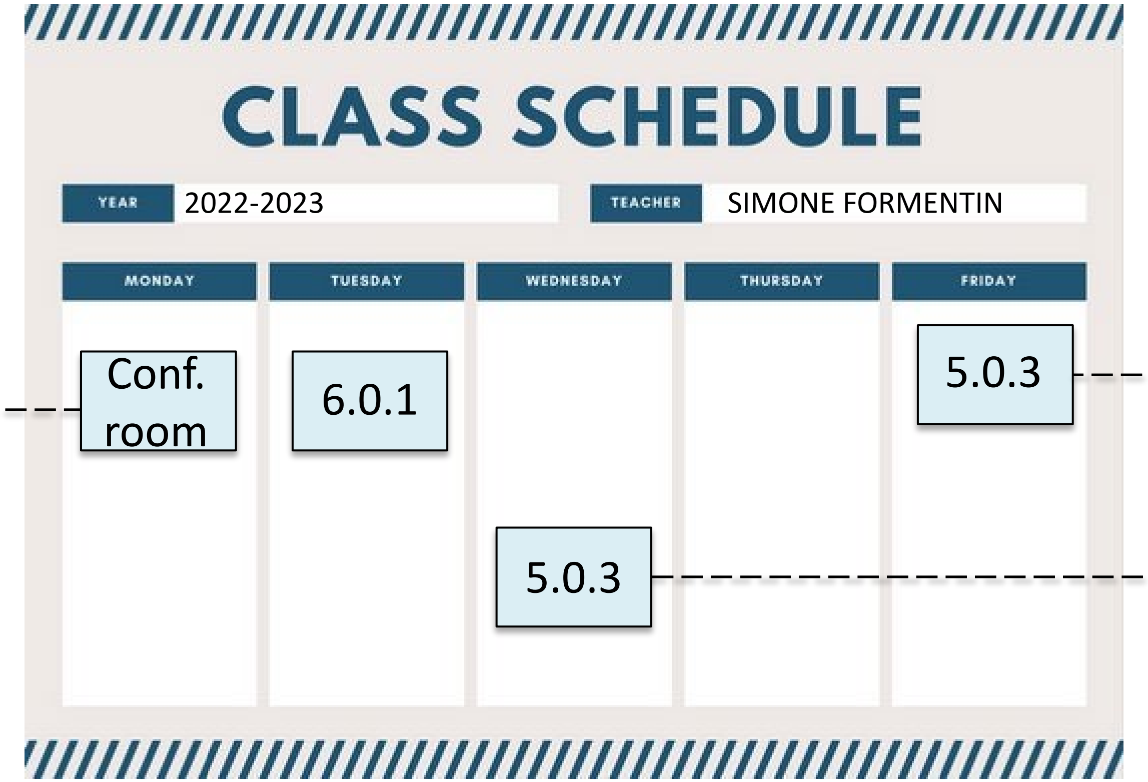
Course organization

From Sept 12 until Oct 25 (Part 2 from Oct 26 until Dec 19)

09:30/
11:00



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10:00

14:30/
16:00

Course organization

- **5 CFU:** 39h lectures/exercises + (optional) 15h lab sessions
- **Lecture notes/exercises/lab sessions/codes/recordings of lectures:** available on WeBeep
- **Prerequisites:** Calculus, Linear Algebra, Statistics
- **Reference textbooks:**
  - ✓ *Learning from data*
    (Amazon, ~$28)
  - ✓ *An introduction to statistical learning with applications in R*
    (freely available at: http://www-bcf.usc.edu/~gareth/ISL/)
- **Evaluation:**
  - ✓ Written exam on SLAS+SIAP (~7 questions, ~140min)
  - ✓ Mid-term call on SLAS (tentative date: Nov 7)

About the labs sessions

Prof. Simone Formentin

Brief History

- Invented in the Netherlands, early 90’s, by **Guido van Rossum**, a dutch programmer

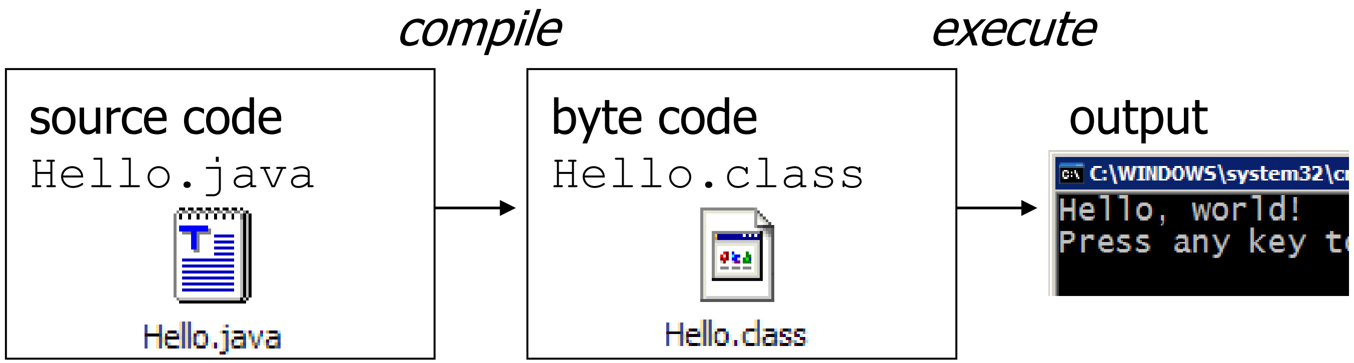
- Named after *Monty Python*, a British surreal comedy group

- Open-source, scalable, object-oriented from the beginning

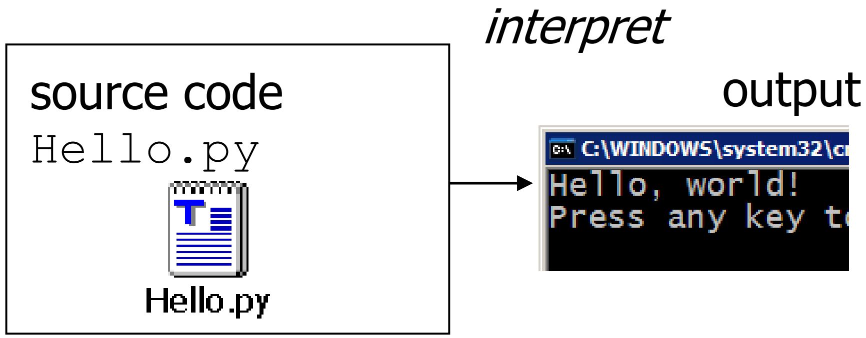
- It is now the **industrial standard for data science**

Compiling and interpreting

- Many languages require you to **compile** (translate) your program into a form that the machine understands.



- Python is instead directly **interpreted** into machine instructions.



Environment set-up

- Anaconda distribution of Python 3 + Jupyter Notebook (not only Python)
- All the notes can be downloaded as .py files and are compatible with any Python IDE or text editor
- http://jupyter.readthedocs.io/en/latest/install.html and follow the instructions (offline version of Jupyter Notebook)
- IMPORTANT: You are not expected to learn Python. However, you can take advantage of the lab sessions to do it!

Libraries (all already included in the Anaconda distribution)

- **Numpy** (Numerical Python package, http://www.numpy.org/)

Library consisting of multidimensional array objects and routines for processing such arrays (similar to Matlab).

- **Pandas** (https://pandas.pydata.org/)

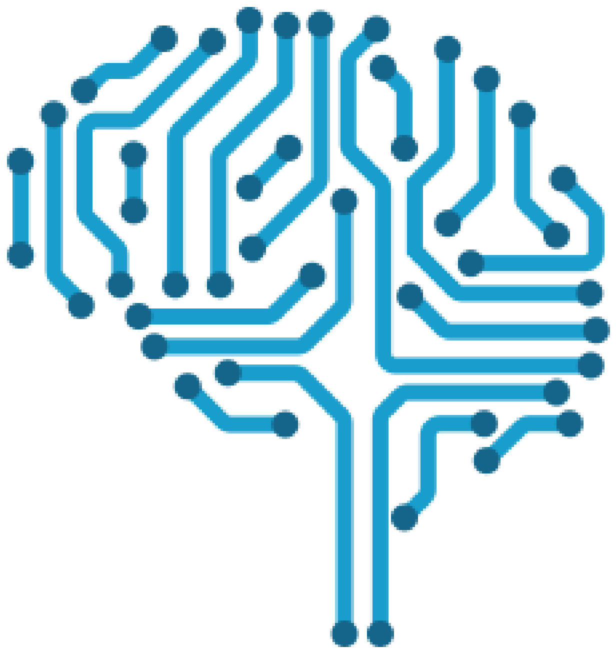
Library of data structures and data analysis tools (advanced Excel-like).

- **Matplotlib** (https://matplotlib.org/)
- **Seaborn** (https://seaborn.pydata.org/)

Libraries for 2D data visualization and statistical plotting, respectively.

- **Scikit-learn** (http://scikit-learn.org/stable/):

Machine learning toolbox.



Introduction to Statistical Learning

Prof. Simone Formentin

LEARNING:

using **Experience**

to gain **Expertise**

What can we learn?

- A task (learn «to do» something)
- A model (a description of the relationship among variables/signals)

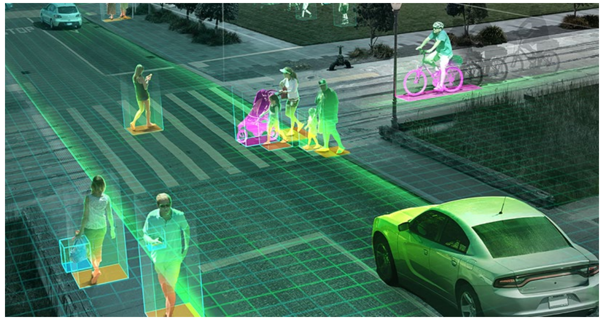
Two types of «learning»

←Human Learning

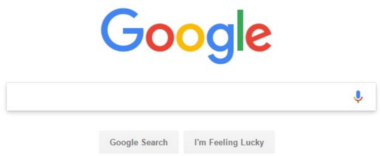
Automated
(machine) learning →

Why to make machines able to learn?

- Some tasks are too **complex to program**
  - ✓ Computer vision: we know to detect objects but not how we do it!



- Some tasks would be too **time-consuming**
  - ✓ Search engines: a human cannot read the entire internet!



- **Adaptivity** and speed of development

But it seems a hard task...is it worth the effort?

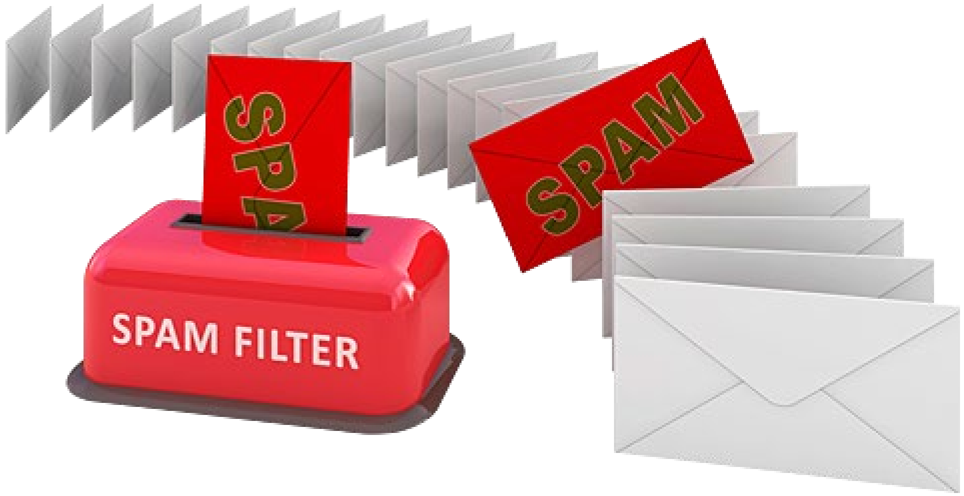
YES!

“ A breakthorugh in machine learning would be worth 10 Microsoft” (Bill Gates)

“Web rankings today are mostly a matter of machine learning” (Prabhakar Raghavan, Vice President of Engineering at Google)

Examples of machine learning applications

Spam filtering:



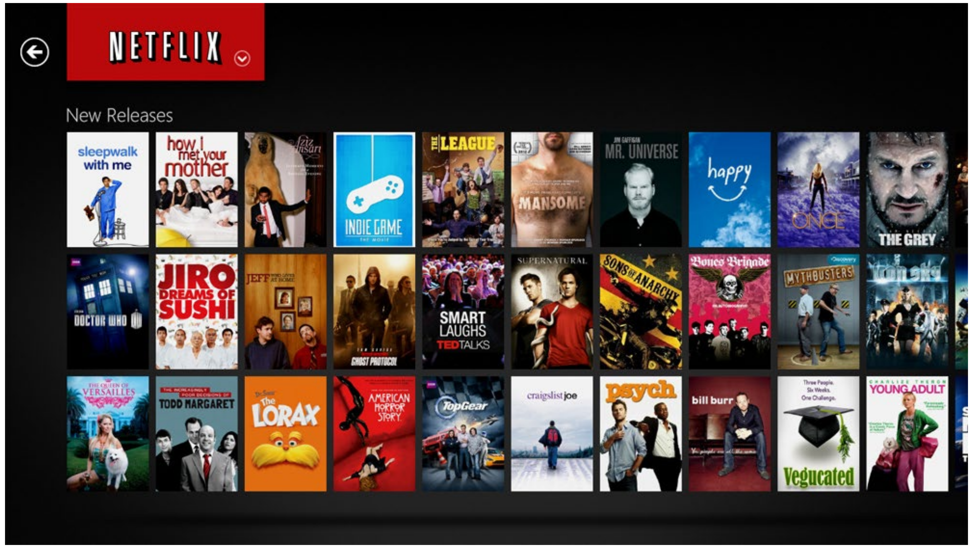
Binary classification: is
this e-mail useful (ham) or spam?

Noisy training data:
Messages previously marked as spam

1° Challenge: Must target errors to user preferences

2° Challenge: Spammers evolve to counter-filter innovations

Movie Rating Prediction:



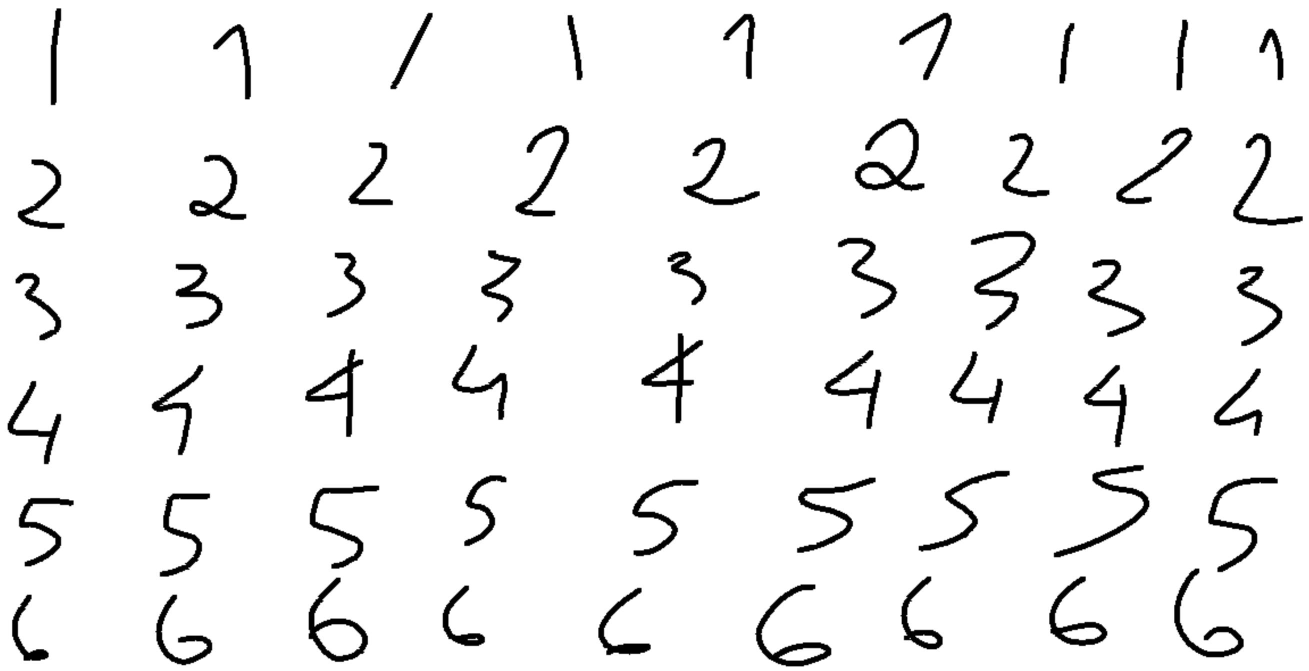
Prediction: What will be the user ranking on a movie?

Noisy training data: rankings of all users on movies

1° Challenge: Must target currents user preferences

2° Challenge: Data is very sparse, i.e., each user rates very few movies!

Handwritten numbers recognition



Examples of machine learning applications

Credit scoring:

age	23 years
annual salary	$30,000
years in residence	1 year
years in job	1 year
current debt	$15,000
...	...

Given the applicant:

Using salary, age, etc., do we approve a loan or not?

Historic data play a key role!

Learning table tennis
(Max Planck Institute and TU Darmstadt):

Crucial in automation systems!

«Classical» industrial robotics

- Very specific task
- Well known environment
- The robot can be «programmed» (expensive and time-consuming)

«New-generation» industrial robotics

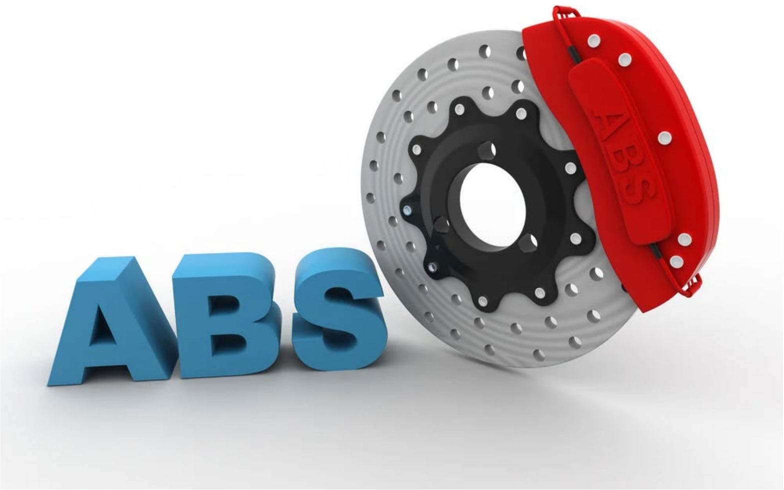
- Complex (multi)-task
- Unknown environment
- The robot cannot be fully «programmed» (learning from experience is necessary)

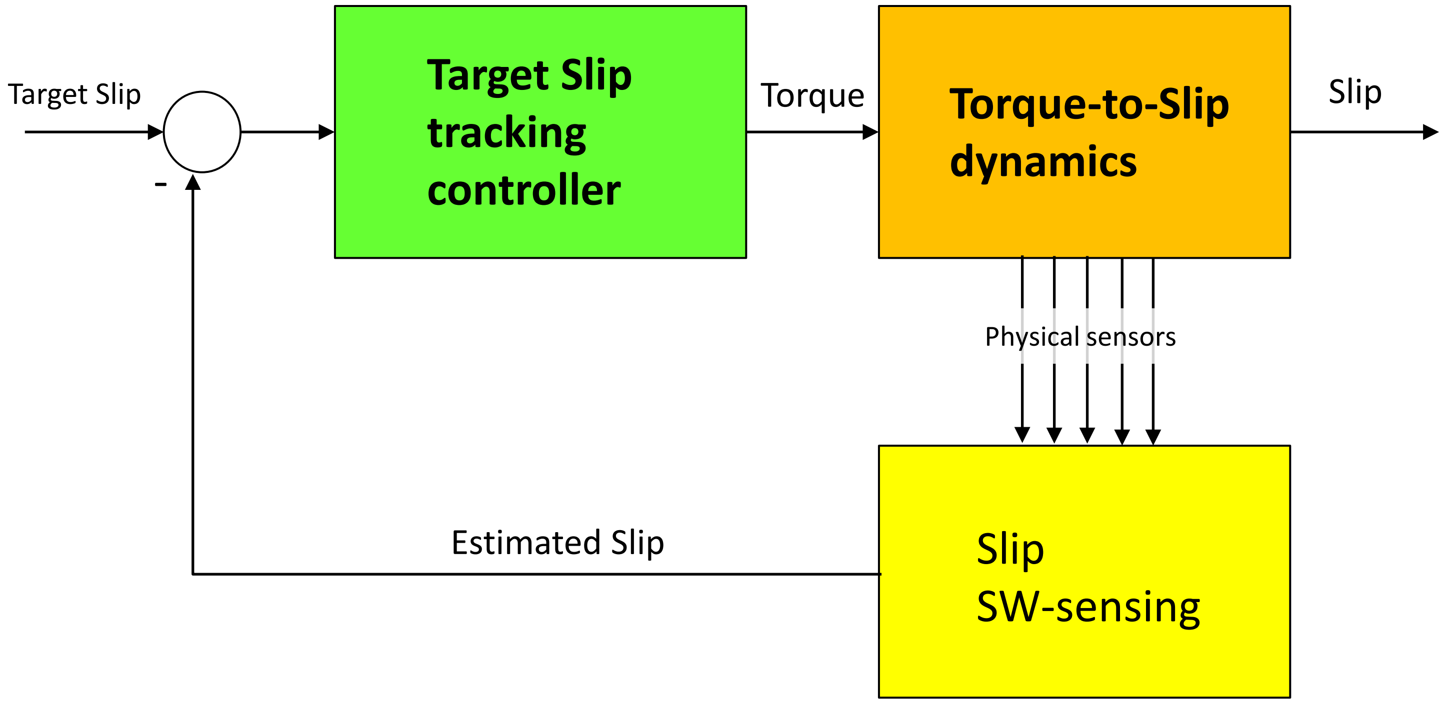
Crucial in automation systems!

Autonomous cars

- Perception problems
- Changing environment
- Strong link with supervision/control

General ML needs of in control problems





- **Black-box** system **modelling**
- **SW-sensing**
- **Auto-tuning** of the controller

Some curiosities

- DeepMind is a company that developed a software which “learns” to play videogames. In 2014, it was acquired by Google (now it is called Google DeepMind) **for 500 M$!!!**

- AlphaGo, developed by Google DeepMind in October 2015 beated the Europe’s Go Champion! (Go is an ancient Chinese game)

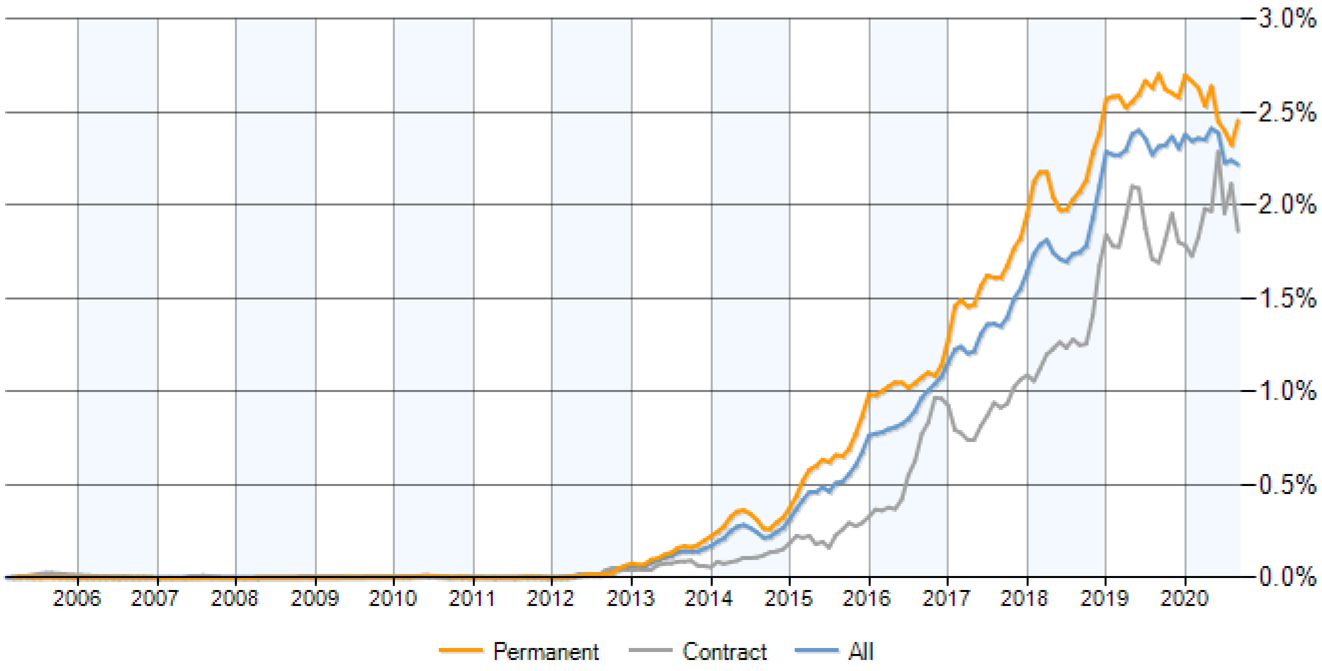
Some curiosities

In **2017**, using data from Bureau of Labor Statistics and other resources, CareerCast ranked the **top 200 professions** in the U.S. according to several key factors, including median salary, expected job growth over the coming years, level of competition, amount of physical work required, safety hazards and amount of stress.

Rank	Title	Job score	Satisfaction	Median Salary
1	Data Scientist	4.8	4.4	$ 110,000
2	DevOps Engineer	4.7	4.2	$ 110,000
3	Data Engineer	4.7	4.3	$ 106,000
5	Analytics Manager	4.6	4.1	$ 112,000
7	Database Admin	4.5	3.8	$ 93,000

Some curiosities

Job postings citing Data Science as a proportion of all IT jobs advertised (source: ITJobsWatch)



Italy, 2019: **one company out of four is hiring a data scientist**!

[Source: Osservatorio Politecnico di Milano]

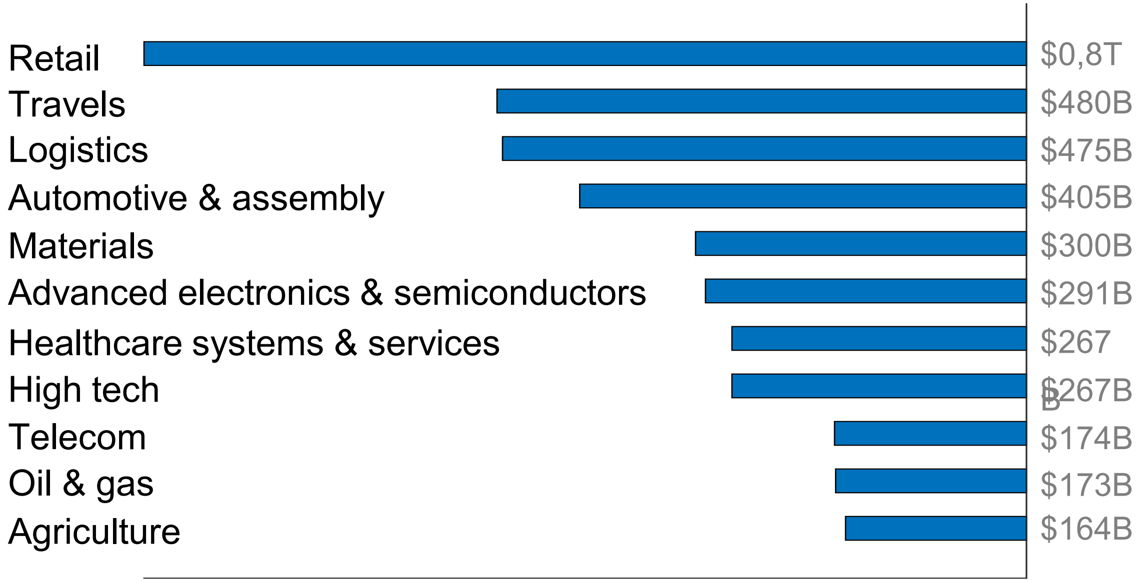
Some curiosities

Expected business value

created by AI until 2030:

$13

Trillions



It is **difficult** to find an industrial sector **that will not benefit** from AI in the near future

LEARNING:

using **Experience**

to gain **Expertise**

LEARNING:

using **data**

to gain **models**

We will focus on Machine Learning!

Statistical

LEARNING:

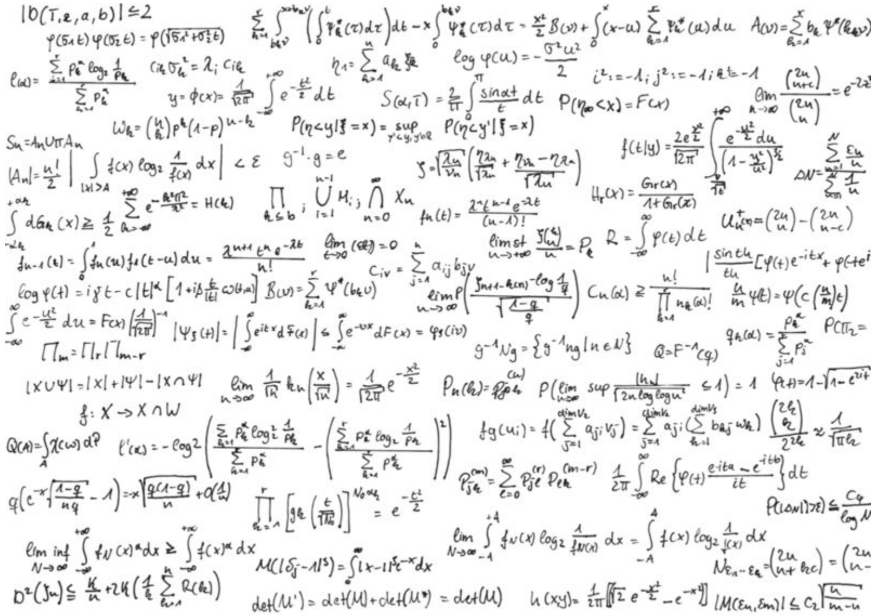
using **data**

to gain **models**

Why «Statistical» Learning?

Finite set of data taken from the system

Learning



Complete rigorous description of the real-world? NO...

The discipline:

Machine learning: "[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed." *(Arthur Samuel, 1959)*

Statistical learning: it is actually machine learning!

Machine learning arose as a subfield of Artificial Intelligence, whereas Statistical learning as a subfield of Statistics.
Now, the distinction has become more and more blurred.

Some basic definitions (to avoid confusion):

Some similar expressions:

Artificial intelligence: “the study of any intelligent agent, i.e., any device that perceives its environment and takes actions that maximize its chance of success at some goal.” *(Poole, Mackworth & Goebel, 1998)*

It is a large set of disciplines that try to “mimic” rationale intelligence in machines. Machine learning obviously plays a key role.

Pattern recognition, **data mining** and **knowledge discovery in databases (KDD):** although they are sometimes referred to as “branches” of machine learning, they are hard to separate from it.

Some basic definitions (to avoid confusion):

House area (feet^2)	**# bedrooms**	**# bathrooms**	**Recently renowed**	**Price** $(1000\$)$
523	1	2	No	115
645	1	3	No	150
708	2	1	No	210
1034	3	3	Si	280
2290	4	4	No	355
2545	4	5	Si	440

A

B

Machine learning

- Predict B given A
- Running software (web site\ mobile app)

Output: Code and program

Data mining

- Houses with 3 bathrooms are more expensive than those with 2 bathrooms of the same size
- Recently renovated houses cost 15% more

Output: Slide deck

supervised

unsupervised

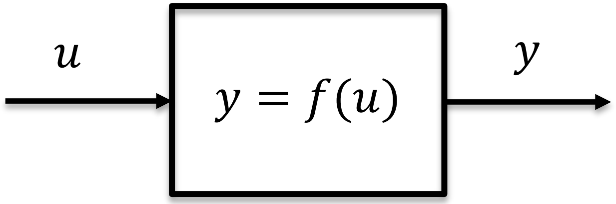
Statistical Learning

semi-supervised

reinforcement

Supervised learning

The science of inferring a function f from a finite set of labeled data



Problem:
Given $\{y_1, y_2, \dots, y_N\}$ corresponding to $\{u_1, u_2, \dots, u_N\}$, find f

Note: when f is a dynamical systems and u and y are time signals, the supervised learning problem is usually known as **System Identification** (see the next course!)

Here: y can be either numerical or categorical variables!

Examples of supervised learning

Pedestrian detection

«Training» dataset

Examples of supervised learning

Pedestrian detection

Labels available!

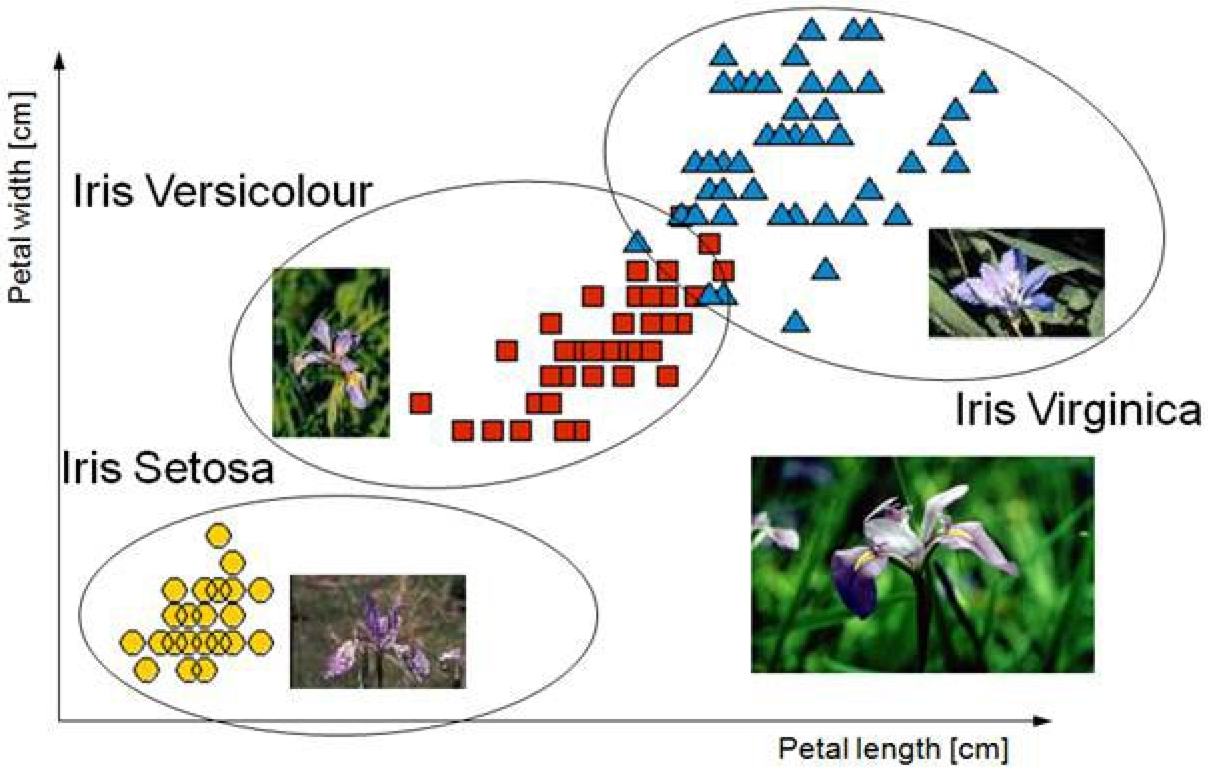
Examples of supervised learning

Pedestrian detection

??

«Test» dataset

Examples of supervised learning



supervised

unsupervised

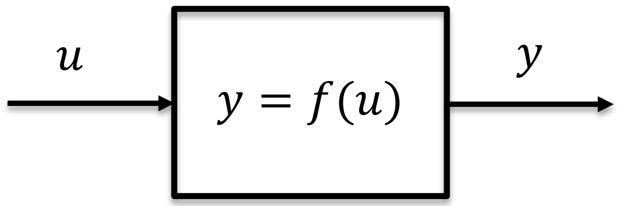
Statistical Learning

semi-supervised

reinforcement

Unsupervised learning

The science of inferring a function f to describe hidden structure from unlabeled data



Problem:
Given $\{u_1, u_2, \dots, u_N\}$, find f

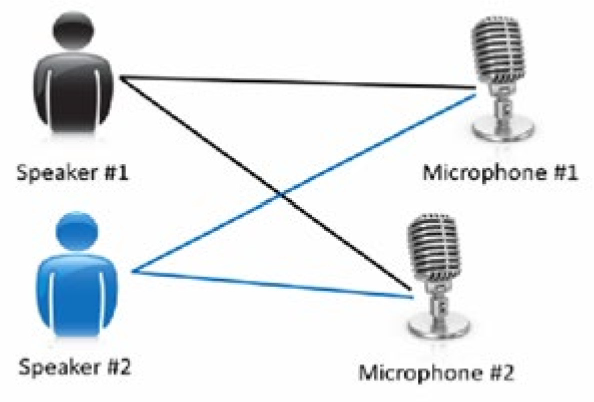
Note: y is NOT an output, but a certain property of the data, i.e., their probability density function or a certain pattern

Example of unsupervised learning

The Cocktail party problem

Lots of overlapping voices (hard to hear what everyone is saying!)

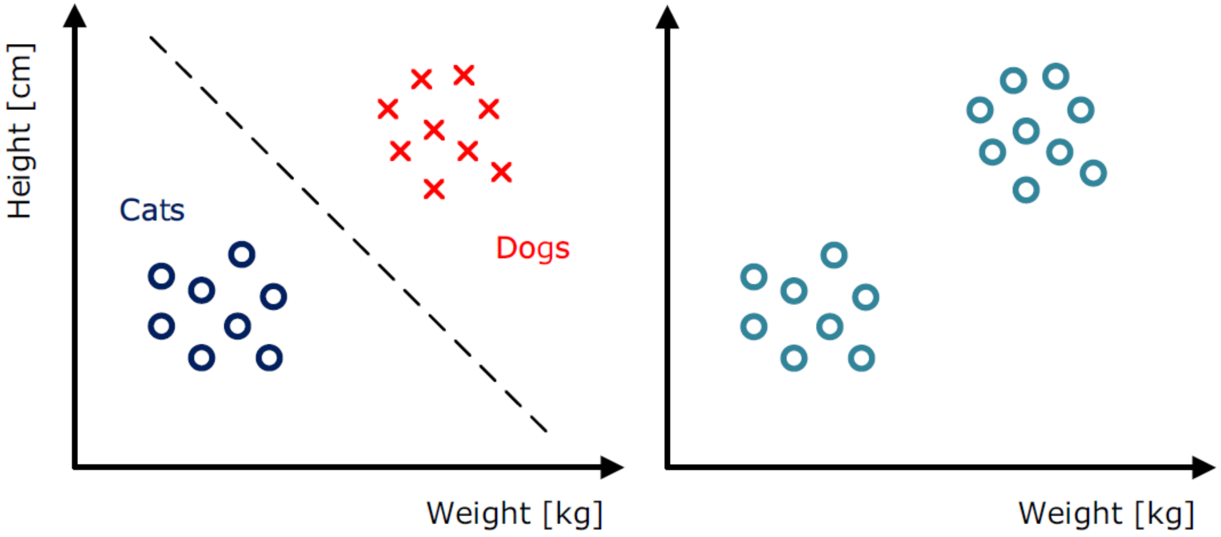
Take 2 people talking with microphones at different distances from the speakers



Problem: to process audio recordings, determine there are 2 audio sources and separate them out

What’s the difference?

Supervised VS Unsupervised



Taxonomy

supervised

unsupervised

Statistical Learning

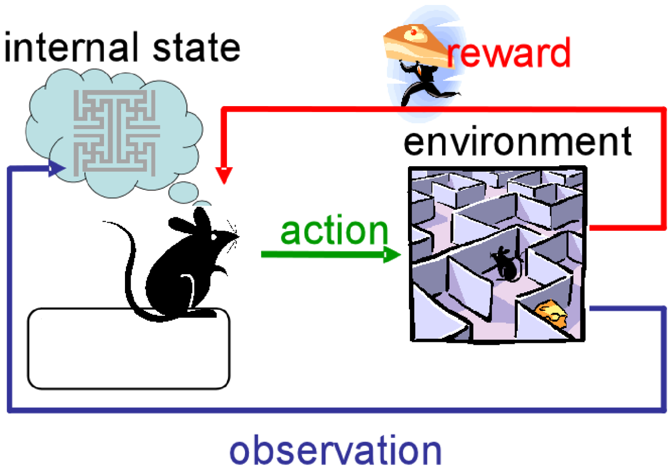
semi-supervised

reinforcement

Semi-supervised learning

Function learning with **mixed** labeled/unlabeled data (NOT IN THIS COURSE)

Reinforcement learning



Learning a task (not a model)…

It's the science of **learning from data a control policy**!
(NOT IN THIS COURSE)

TAKE HOME LESSONS

- **Automated learning** is the key ingredient of future ICT (not only automation and control)
- It's definitely worthwhile to learn data science in general! (very good jobs, it makes the difference in a business…)
- Machine learning, data mining, patter recognition…several names for similar stuff!
- In this course, 4 main topics:
  - ✓ Mathematical foundations of learning theory
  - ✓ Supervised learning (dynamical models in SIAP)
  - ✓ Unsupervised learning
  - ✓ Learning with time series

Talking about «learning»...